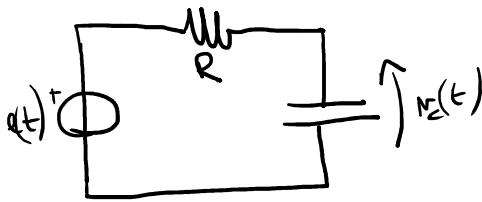


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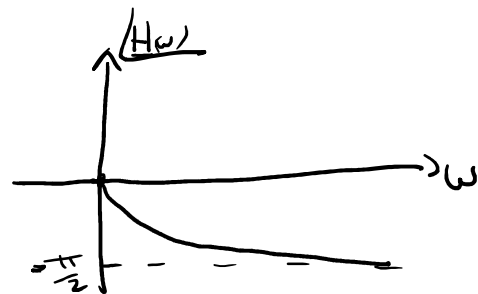
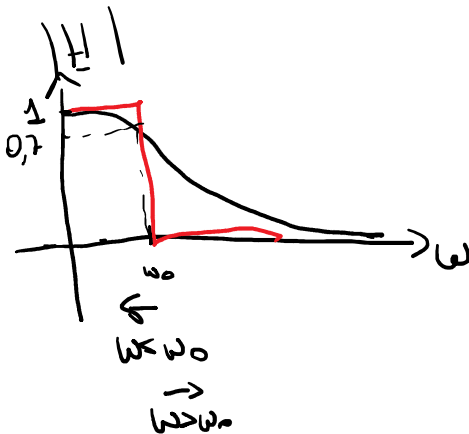
$$\underline{H}(\omega) = \frac{\underline{V}_c(\omega)}{\underline{E}(\omega)}$$

$$\underline{V}_c = \underline{E} \cdot \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{1}{j\omega CR + 1} \underline{E}$$

$$\underline{H} = \frac{1}{1 + j\omega CR} \rightarrow |\underline{H}| = \frac{1}{\sqrt{1 + \omega^2 C^2 R^2}} \quad \boxed{\omega_0 = \frac{1}{RC}}$$

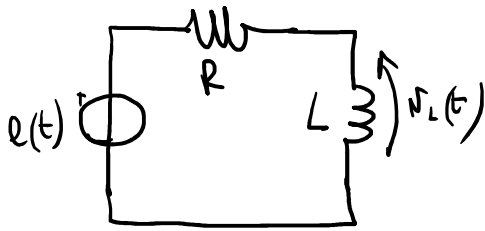
$$\rightarrow \underline{H} = \frac{1}{1 + j\omega CR} \cdot \frac{1 - j\omega CR}{1 - j\omega CR} =$$

$$= \frac{1 - j\omega CR}{1 + \omega^2 C^2 R^2} \rightarrow \angle \underline{H} = \arctan\left(\frac{-\omega CR}{1}\right)$$



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$$\underline{H}(\omega) = \frac{\underline{V}_L}{\underline{E}}$$

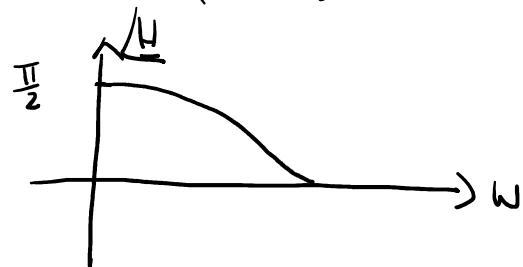
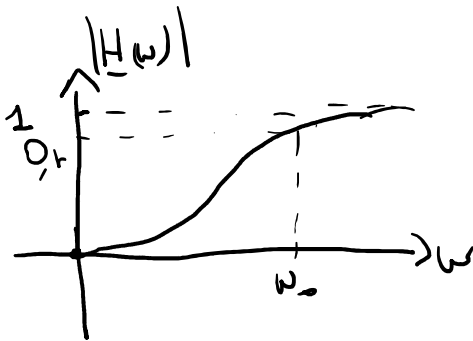
$$\underline{V}_L = \frac{j\omega L}{R + j\omega L} \underline{E} \rightarrow \underline{H}(\omega) = \frac{j\omega L/R}{1 + j\omega L/R}$$

$$|\underline{H}| = \frac{\omega L/R}{\sqrt{1 + (\omega L/R)^2}}$$

$$\omega_0 = \frac{R}{L}$$

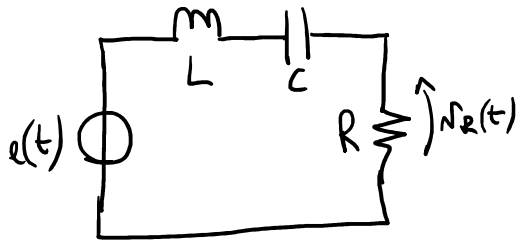
$$\underline{H} = \frac{j\omega L}{R + j\omega L} \cdot \frac{R - j\omega L}{R - j\omega L} = \frac{j\omega L R + \omega^2 L^2}{R^2 + \omega^2 L^2}$$

$$\angle \underline{H} = \tan^{-1} \left(\frac{\omega L R}{\omega^2 L^2} \right)$$



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$$\underline{H}(\omega) = \frac{\underline{V}_R}{\underline{E}}$$

$$\underline{V}_R = \frac{R}{R + j\omega L - \frac{j}{\omega C}} \quad \underline{E} = \frac{R}{R + j(\omega L - \frac{1}{\omega C})}$$

$$|\underline{H}| = \frac{R}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\underline{H} = \frac{R}{R + j(\omega L - \frac{1}{\omega C})} \cdot \frac{R - j(\omega L - \frac{1}{\omega C})}{R - j(\omega L - \frac{1}{\omega C})} =$$

$$= \frac{R^2 - jR(\omega L - \frac{1}{\omega C})}{R^2 + \omega L - \frac{1}{\omega C}} \rightarrow \angle \underline{H} = \tan^{-1} \left(\frac{-R(\omega L - \frac{1}{\omega C})}{R^2} \right)$$

